Bug Reports — HW1: Smart Gate Controller

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Defects found during black-box functional testing of the Smart Gate Controller (27 test cases: 23 passed, 3 failed, 1 blocked). Build result: Rejected for release.

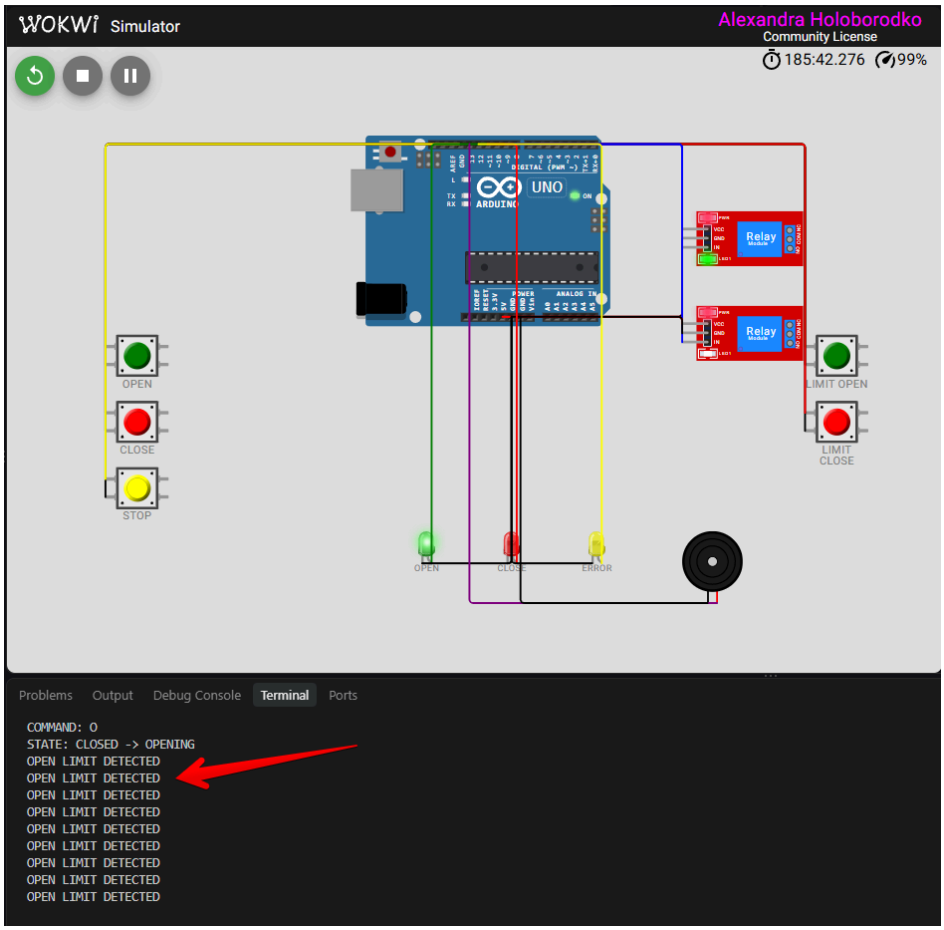
Environment: Wokwi, Arduino Uno, Build_1

Summary

ID	Title	Severity
Bug_1	State does not change to OPEN after LIMIT_OPEN is activated	Critical
Bug_2	ERROR is not raised after >10 s in OPENING state without OPEN limit	High
Bug_3	ERROR is not raised when both limits are activated at the same time	High

Bug_1 — State does not change to OPEN after LIMIT_OPEN is activated

Requirement	R6, R15
Precondition	State = OPENING; OPEN relay = ON; CLOSE relay = OFF; OPEN LED = ON
Steps	1. Start the simulation and send OPEN (State = OPENING). 2. Verify State = OPENING. 3. Activate LIMIT_OPEN. 4. Observe state, relays and indication.
Expected	State = OPEN OPEN relay = OFF; CLOSE relay = OFF Indication corresponds to OPEN
Actual	Terminal shows "OPEN LIMIT DETECTED", but the state does not become OPEN. OPEN LED = ON; OPEN relay = OFF; CLOSE relay = OFF (relays and LED behave as expected). TC-04 (CLOSE from OPEN) is blocked because the OPEN state is unreachable.
Reproducibility	5/5
Environment	Wokwi (simulation), Arduino Uno; Smart Gate Controller, Build_1; black-box; tested 24–26.09.2026
Evidence / Notes	Failed test: TC-03. Blocks TC-04. Screenshot below
Severity	Critical



Bug_2 — ERROR is not raised after >10 s in OPENING state without OPEN limit

Requirement	R9
Precondition	State = OPENING; OPEN relay = ON; LIMIT_OPEN not activated
Steps	1. Send OPEN (State = OPENING). 2. Do not activate LIMIT_OPEN. 3. Wait >10 s and observe state, relays and indication.
Expected	ERROR; relays OFF; error indication (ERROR LED ON)
Actual	No ERROR. State stays OPENING for more than 10 s; OPEN relay stays ON.
Reproducibility	5/5
Environment	Wokwi (simulation), Arduino Uno; Smart Gate Controller, Build_1; black-box; tested 24–26.09.2026
Evidence / Notes	Failed test: TC-13. The same check for CLOSING (TC-14) passes, so the timeout is missing only for OPENING.
Severity	High

Bug_3 — ERROR is not raised when both limits are activated at the same time

Requirement	R10
Precondition	Any valid state (tested in STOPPED); both relays OFF
Steps	1. Bring the controller to STOPPED. 2. Activate LIMIT_OPEN and LIMIT_CLOSE together. 3. Observe state, relays and indication.
Expected	ERROR; relays OFF; error indication (ERROR LED ON)
Actual	No ERROR; the contradictory sensor state is not detected.
Reproducibility	5/5
Environment	Wokwi (simulation), Arduino Uno; Smart Gate Controller, Build_1; black-box; tested 24–26.09.2026
Evidence / Notes	Failed test: TC-15.
Severity	High